Codebook for QUETRANSPORT

A general quantitative spatial toolkit for transport appraisal

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Abstract

QUETRANSPORT implements a deliberately streamlined version of the quantitative urban model in Ahlfeldt, Redding, Sturm, and Wolf (2015, ARSW) for an arbitrary number of spatial units. It takes standardized spatial data and baseline and policy travel-time matrices, inverts local fundamentals conditional on chosen parameters, solves closed- and open-city counterfactual equilibria, and evaluates a fixed-distribution accounting benchmark. Each is also evaluated after switching off productivity and amenity spillovers. This codebook states the assumptions and equations implemented in MATLAB, shows that the equilibrium has as many equations as unknown target variables, documents every inverted primitive, and distinguishes the common observed floor-space rent, endogenous use-specific bid rents, and annual land rent. Structural estimation is outside the toolkit.

Contents

1 Purpose, scope, and sequence

The toolkit answers a conditional question: given a quantitative spatial model, a baseline city, parameter values, and a transport change, what new spatial equilibrium does the model imply? It does not estimate the structural parameters. The user chooses variable mappings, parameters, shocks, numerical controls, and city closure in the root file `project_config.yaml`.

The workflow has four economic steps:

1. specify parameter values and assemble mutually consistent spatial inputs;
2. invert the baseline fundamentals that rationalize the observed allocation;
3. change travel times and, optionally, productivity, amenity, or development fundamentals;
4. solve the baseline and policy equilibria under the same closure, evaluate the fixed-distribution benchmark, and report local, aggregate, welfare, and land-rent effects.

Every location vector has length N. Both travel-time matrices are $N \times N$ and use the same location ordering as the standardized table and shapefile. GRID may construct regular squares or hexagons, or use `cell_geometry: original`. In the latter case each supplied source polygon is retained as a model location, the cell-size parameter is ignored, and the same polygons determine routing centroids, MATLAB locations, shock interpolation targets, and reporting maps. A configured unique source ID may be retained as the canonical location identifier.

2 Economic environment and assumptions

Locations are indexed by residence i and workplace j. The model has the following maintained assumptions.

A1. There is a continuum of workers. Each supplies one unit of labour and chooses a residence–workplace pair.

A2. Workers have Frechet-distributed idiosyncratic preferences. This produces a smooth gravity equation for commuting and a closed-form expected-utility index.

A3. Commuting reduces the effective wage exponentially in bilateral travel time. Travel times are measured in the unit used to calibrate κ.

A4. A homogeneous, freely traded final good is the numeraire. Firms are perfectly competitive and use labour and commercial floor space under constant returns to scale.

A5. Residents spend share β of income on the numeraire and share $1-\beta$ on residential floor space.

A6. Geographic land is fixed. A reduced-form developer technology converts land and the non-land development input into floor space.

A7. Productivity and amenities may depend on surrounding employment and population densities. Setting the corresponding spillover elasticity to zero turns that feedback off.

A8. In a closed city total population is fixed and utility adjusts. In an open city utility is fixed at the outside option and total population adjusts.

A9. The baseline inversion is conditional on parameter values and observed quantities and rents. It is an exact model-based quantification, not an estimation exercise.

3 Workers, commuting, and welfare

Let B_i be total amenity, Q_i residential floor-space rent, w_j the wage, and τ_{ij} travel time. Define bilateral attractiveness

$$\Phi_{ij} = e^{-\epsilon\kappa\tau_{ij}} B_i^{\epsilon} Q_i^{-(1-\beta)\epsilon} w_j^{\epsilon}, \qquad \Phi = \sum_i \sum_j \Phi_{ij}. \tag{1}$$

The unconditional probability of residing in i and working in j is

$$\pi_{ij} = \frac{\Phi_{ij}}{\Phi}. \tag{2}$$

For total city population H,

$$H_{Ri} = H \sum_j \pi_{ij}, \qquad H_{Mj} = H \sum_i \pi_{ij}, \qquad \sum_i H_{Ri} = \sum_j H_{Mj} = H. \tag{3}$$

Conditional commuting probabilities and resident income are

$$\pi_{j|i} = \frac{\Phi_{ij}}{\sum_s \Phi_{is}}, \qquad \bar{w}_i = \sum_j \pi_{j|i} w_j, \qquad v_i = H_{Ri}\bar{w}_i. \tag{4}$$

Thus v_i is total labour income of residents of i, which is the object used in residential floor-space demand. Expected utility is

$$U = \gamma \Phi^{1/\epsilon}, \qquad \gamma = \Gamma\left(\frac{\epsilon - 1}{\epsilon}\right). \tag{5}$$

4 Production, floor space, and land

Total floor space in location i is L_i. Share θ_i is commercial, so $L_{Mi} = \theta_i L_i$ and $L_{Ri} = (1 - \theta_i)L_i$. Production is

$$Y_i = A_i H_{Mi}^{\alpha} L_{Mi}^{1-\alpha}. \tag{6}$$

Perfect competition gives the labour and commercial-floor-space conditions

$$w_i = \frac{\alpha Y_i}{H_{Mi}}, \qquad q_i = \frac{(1-\alpha)Y_i}{L_{Mi}}, \tag{7}$$

where q_i is commercial floor-space rent. Residential market clearing is

$$Q_i L_{Ri} = (1 - \beta) v_i, \tag{8}$$

where Q_i is residential floor-space rent.

Let K_i denote geographic land and let μ be the non-land share in development. The reduced-form floor-space stock is

$$L_i = \varphi_i K_i^{1-\mu}. \tag{9}$$

The configured values are $\mu = 0.75$ and $1 - \mu = 0.25$. Hence land enters (9) with exponent 0.25, not 0.75.

4.1 Mixed and specialized locations

The active support of productivity and amenity determines whether a location can host firms, residents, or both. For a mixed location, no arbitrage between uses implies

$$q_i = Q_i = r_i^F, \tag{10}$$

and combining (7)–(8) yields

$$r_i^F = \frac{(1-\alpha)Y_i + (1-\beta)v_i}{L_i}, \tag{11}$$

$$\theta_i = \frac{(1-\alpha)Y_i}{r_i^F L_i}. \tag{12}$$

In a commercially specialized location $\theta_i = 1$ and the commercial condition determines q_i. In a residentially specialized location $\theta_i = 0$ and the residential condition determines Q_i.

4.2 One observed rent and two endogenous bid rents

The standardized input supplies one common baseline floor-space rent, R_i^{obs}. The inversion uses this same observation in both the commercial and residential demand equations. It is neither land rent nor a regulatory wedge. This is a transparent maintained restriction on baseline data.

The counterfactual model nevertheless retains two endogenous bid rents: commercial q_i and residential Q_i. Economic demand differs across the two uses. In a mixed-use location the no-arbitrage condition (10) imposes equality; at a specialized corner only the active use's bid rent clears its market, so q_i and Q_i need not coincide. Annual land rent $\mathcal{R}_i$ remains the separate developer residual in (13).

4.3 Floor-space rents, annual land rent, and land value

The model contains three distinct concepts:

1. q_i: price per unit of commercial floor space;
2. Q_i: price per unit of residential floor space;
3. $\mathcal{R}_i$: annual rent paid to the fixed factor land.

Under the constant-returns developer problem, Euler's theorem assigns the land share $1-\mu$ of gross floor-space revenue to land:

$$\boxed{\mathcal{R}_i = (1-\mu)\left(q_i L_{Mi} + Q_i L_{Ri}\right).} \tag{13}$$

This is the object computed by `qt_write_closure_results.m` and reported as `TotalLandRentPct`. It is a *derived endogenous equilibrium outcome*: rents and land-use shares change in equilibrium, and (13) maps those outcomes into the developer residual. It is not a floor-space rent and is not their unweighted sum. A capitalized land asset value would additionally require a capitalization rate (and expectations about future rents), so the toolkit does not label annual land rent as land value.

Aggregate annual land rent is

$$\mathcal{R} = \sum_i \mathcal{R}_i, \qquad \%\Delta\mathcal{R} = 100\left(\frac{\mathcal{R}^1}{\mathcal{R}^0} - 1\right). \tag{14}$$

5 Endogenous productivity and amenity

Employment-density and residential-density spillovers are

$$\Upsilon_i = \sum_s e^{-\delta\tau_{is}} \frac{H_{Ms}}{K_s}, \qquad A_i = a_i \Upsilon_i^{\lambda}, \tag{15}$$

$$\Omega_i = \sum_r e^{-\rho\tau_{ir}} \frac{H_{Rr}}{K_r}, \qquad B_i = b_i \Omega_i^{\eta}. \tag{16}$$

Here a_i and b_i are fundamentals held fixed across a pure transport counterfactual; A_i and B_i are total productivity and amenity, which respond endogenously. Setting $\lambda = 0$ or $\eta = 0$ makes the corresponding total object equal to its fundamental.

6 Objects and their status

Table 1: Structural, supplied, inverted, and endogenous objects

Object	Meaning and status
Parameters fixed by the user	
α, β	Labour share in production and numeraire-consumption share.
ϵ, κ	Commuting-choice elasticity and time-cost coefficient.
λ, δ	Productivity-spillover elasticity and spatial decay.
η, ρ	Amenity-spillover elasticity and spatial decay.
$\mu, 1-\mu$	Non-land and land shares in development.
Baseline data or externally supplied objects	
K_i	Geographic land area.
H_{Ri}^{obs}	Resident population.
H_{Mi}^{obs}	Workplace employment, rescaled to equal population in aggregate.
R_i^{obs}	One common observed baseline floor-space rent, used in both inversion conditions.
τ^0, τ^1	Baseline and counterfactual bilateral travel times.
Primitives recovered by baseline inversion	
a_i, b_i	Productivity and amenity fundamentals net of endogenous spillovers.
φ_i	Structural-density/floor-supply fundamental.
$\bar{U}$	Baseline utility used as outside utility in the open-city model.
Adjusted baseline objects recovered during inversion	
A_i, B_i	Total productivity and amenity that rationalize the two employment margins.
$w_i, \bar{w}_i, v_i$	Workplace wage, expected wage, and resident total income.
$L_i, L_{Mi}, L_{Ri}, \theta_i$	Total and use-specific floor space and commercial share.
Endogenous counterfactual equilibrium outcomes	
π_{ij}, H_{Ri}, H_{Mi}	Commuting probabilities and residential/workplace employment.
A_i, B_i, Y_i	Total productivity, total amenity, and output.
w_i, q_i, Q_i, θ_i	Wage, two endogenous use-specific bid rents, and commercial floor-space share.
U, H	Utility and population; which is fixed depends on closure.
$\mathcal{R}_i, \mathcal{R}$	Annual local and aggregate land rent from the developer residual.

7 Baseline inversion

The inversion conditions on $\{R^{obs}, H_M^{obs}, H_R^{obs}, K, \tau^0\}$ and the parameter vector. Employment is first rescaled so $\sum_i H_{Mi}^{obs} = \sum_i H_{Ri}^{obs} = H$.

7.1 Adjusted productivity, amenity, and wages

Combining the firm's two first-order conditions eliminates commercial floor space and gives the wage equation implemented in `cmodexog.m`:

$$w_i = \alpha A_i^{1/\alpha} \left(\frac{1-\alpha}{R_i^{obs}} \right)^{(1-\alpha)/\alpha}. \tag{17}$$

Given guesses for A and B, equations (1)–(3) produce predicted employment margins $\widehat{H}_M$ and $\widehat{H}_R$. The multiplicative inversion mappings are

$$A_i^* = A_i \left(\frac{H_{Mi}^{obs}}{\widehat{H}_{Mi}} \right)^{1/\epsilon}, \qquad B_i^* = B_i \left(\frac{H_{Ri}^{obs}}{\widehat{H}_{Ri}} \right)^{1/\epsilon}. \tag{18}$$

The code damps these mappings, normalizes the geometric mean of positive A_i to one, and rescales B so $\Phi = H$. At convergence,

$$\widehat{H}_{Mi} = H^{obs}_{Mi}, \qquad \widehat{H}_{Ri} = H^{obs}_{Ri} \quad \forall i. \tag{19}$$

The productivity normalization fixes the nominal scale; amenity scaling adopts the ARSW utility normalization.

7.2 Floor space and structural density

After convergence, all remaining floor-space objects are recovered directly:

$$L_{Mi} = \left(\frac{(1-\alpha)A_i}{R^{obs}_i} \right)^{1/\alpha} H^{obs}_{Mi}, \tag{20}$$

$$L_{Ri} = \frac{(1-\beta)v_i}{R^{obs}_i}, \tag{21}$$

$$L_i = L_{Mi} + L_{Ri}, \qquad \theta_i = \frac{L_{Mi}}{L_i}, \tag{22}$$

$$\varphi_i = \frac{L_i}{K_i^{1-\mu}}. \tag{23}$$

The same R^{obs}_i enters both use-specific equations, but it must not be confused with land rent. This is the toolkit's maintained one-rent baseline restriction; it does not impose equal commercial and residential demand.

7.3 Productivity and amenity fundamentals

Using observed employment densities and baseline travel times, compute Υ^0_i and Ω^0_i from (16). The fundamentals held fixed in a pure transport experiment are

$$a_i = \frac{A_i}{(\Upsilon^0_i)^\lambda}, \qquad b_i = \frac{B_i}{(\Omega^0_i)^\eta}. \tag{24}$$

Finally, baseline outside utility follows from (5):

$$\bar{U} = \gamma(\Phi^0)^{1/\epsilon}. \tag{25}$$

Equations (17), (4), (20)–(25) cover every primitive and adjusted object saved by the inversion.

Algorithm 1 Baseline inversion

Require: $R^{obs}, H^{obs}_M, H^{obs}_R, K, \tau^0$ and parameters

Rescale employment so both observed margins sum to H
Initialize positive A and B
while either employment margin fails (19) **do**
    Compute wages from (17)
    Compute Φ_{ij}, π_{ij}, $\widehat{H}_M$, and $\widehat{H}_R$
    Update A and B using (18) and damping
    Normalize A and scale B consistently with the utility normalization
end while
Compute conditional commuting probabilities and resident income
Recover $L_M, L_R, L, \theta, \varphi$ from (20)–(23)
Recover a, b from (24) and $\bar{U}$ from (25)

Ensure: Baseline primitives, adjusted objects, outcomes, and diagnostics

8 Counterfactual equilibrium

A pure transport experiment changes τ^0 to τ^1 while holding $\{a, b, \varphi, K\}$ and all parameters fixed. The user may instead combine the transport change with multiplicative local changes $\widehat{a}_i$, $\widehat{b}_i$, and $\widehat{\varphi}_i$:

$$a_i^1 = a_i^0 \widehat{a}_i, \qquad b_i^1 = b_i^0 \widehat{b}_i, \qquad \varphi_i^1 = \varphi_i^0 \widehat{\varphi}_i. \tag{26}$$

A hat of one leaves that fundamental unchanged; for example, 1.10 raises it by ten percent. Conditional on target variables, equations (1)–(16) explicitly construct all other outcomes.

The optional user input is a polygon shapefile or GeoPackage in arbitrary spatial units. It need not match the endogenous model grid. Let G_i denote a target cell, P_s a source policy polygon, and $|\cdot|$ polygon area in a projected coordinate system. GRID maps any source hat $\widehat{x}_s$ into

$$\widehat{x}_i = 1 + \sum_s \frac{|G_i \cap P_s|}{|G_i|} (\widehat{x}_s - 1). \tag{27}$$

Uncovered target area therefore contributes the neutral value one. When source polygons form a complete non-overlapping partition, equation (27) is the standard intersection-area-weighted mean. Overlapping policy polygons are rejected to prevent double counting. The standardized model-order table contains `productivity_hat`, `amenity_hat`, and `structural_density_hat`. Before MATLAB runs, the reporting layer maps all three interpolated percentage changes, overlays the original policy-polygon boundaries and transport innovation, and saves both PDF and PNG versions. This provides a direct visual check of (27) and the transport intervention. If no policy geography is configured, all three hats are one.

The current one-rent specification has no regulatory-wedge primitive; a wedge shock is therefore rejected rather than silently ignored. Adding one would require specifying how the wedge enters developer returns or land-use arbitrage and adding the corresponding inversion and equilibrium conditions.

8.1 Equations and unknowns

For N active locations, the closed-city fixed point targets

$$\underbrace{\{w_i, q_i, Q_i, \theta_i\}_{i=1}^N}_{4N \text{ unknowns}}. \tag{28}$$

There are exactly $4N$ corresponding equilibrium conditions:

1. N labour first-order conditions, $w_i = \alpha Y_i / H_{Mi}$;
2. N commercial-rent conditions, using $q_i = (1 - \alpha) Y_i / (\theta_i L_i)$ in commercial locations and (11) in mixed locations;
3. N residential-rent conditions, using $Q_i = (1 - \beta) v_i / [(1 - \theta_i) L_i]$ in residential locations and (11) in mixed locations;
4. N land-use conditions: (12) in mixed locations and the appropriate specialization restriction $\theta_i \in \{0, 1\}$ elsewhere.

The commuting probabilities, employment margins, A, B, Y, and v are not omitted equations. They are explicit mappings from the four target vectors through (1)–(16); equivalently, one could list them as additional unknowns together with the same number of defining equations.

For the open city, total population H is an additional scalar unknown and

$$U = \bar{U} \tag{29}$$

is the additional scalar equation. The open-city system therefore has $4N + 1$ target unknowns and $4N + 1$ target equations.

8.2 Scenario design: what adjusts and what remains fixed

The toolkit reports three conceptually different responses to the same transport intervention. The closed and open cities are market-clearing spatial equilibria. The fixed-distribution case is a deliberately partial-equilibrium accounting benchmark. Table 2 summarizes their identifying restrictions.

Table 2: Scenario comparison for a pure transport counterfactual

	Closed city	Open city	Fixed distribution
City population	Fixed at baseline	Adjusts endogenously	Fixed at baseline
Expected utility	Adjusts endogenously	Fixed at outside option	Accounting measure adjusts
OD assignments	Adjust endogenously	Adjust endogenously	Fixed at baseline
Local population and jobs	Adjust endogenously	Adjust endogenously	Fixed at baseline
Wages and output	Adjust endogenously	Adjust endogenously	Recomputed at fixed inputs
Floor-space rents and land use	Adjust endogenously	Adjust endogenously	Fixed at baseline
Transport-mediated spillovers	Recomputed	Recomputed	Recomputed at fixed densities
Market clearing	Yes	Yes	No: accounting benchmark

The scenarios answer distinct questions. The *closed city* asks how a fixed set of workers sorts across residences and workplaces once commuting conditions change. Utility is the natural worker-welfare statistic; GDP and land rent include the effects of relocation, rent adjustment, and endogenous spillovers. The *open city* asks how many workers the city attracts when utility is tied to an outside option. Its near-zero reported utility change is therefore a model restriction, not an absence of benefits. Benefits are capitalized into population, output, rents, and land rent. The *fixed-distribution* case asks what the original commuters would gain if they retained their baseline OD assignments and local density distributions. It includes direct commuting gains and travel-time-mediated productivity and amenity gains, but excludes relocation, resorting, migration, rent capitalization, and land-use adjustment.

8.3 Closed city

The closed city fixes $H = H^0$. The equilibrium mapping is

$$L_i = \varphi_i K_i^{1-\mu}, \tag{30}$$

$$\pi_{ij} = \pi_{ij}(B, Q, w, \tau), \qquad (H_R, H_M) = H\,\mathrm{margins}(\pi), \tag{31}$$

$$A_i = a_i \Upsilon_i(H_M, \tau)^{\lambda}, \qquad B_i = b_i \Omega_i(H_R, \tau)^{\eta}, \tag{32}$$

$$Y_i = A_i H_{Mi}^{\alpha}(\theta_i L_i)^{1-\alpha}, \qquad v_i = H_{Ri} \sum_j \pi_{j|i} w_j, \tag{33}$$

followed by the $4N$ target conditions above. Utility from (5) is the aggregate welfare outcome.

8.4 Open city

The open city fixes $\bar{U}$ at the inverted baseline level. In addition to the closed-city mapping, the code uses

$$H^* = H\left(\frac{U}{\bar{U}}\right)^{\epsilon} \tag{34}$$

and damps population toward H^*. If current city utility exceeds the outside option, population rises. Convergence requires both all local target equations and (29).

Algorithm 2 Closed-city equilibrium

Require: $a, b, \varphi, K, \tau, H$ and parameters

Compute $L = \varphi K^{1-\mu}$ and initialize w, q, Q, θ

while any target vector has not converged **do**

  Construct A and B from current employment densities

  Construct commuting probabilities and allocate fixed H

  Recompute A, B, Y, conditional probabilities, and v

  Predict w, q, Q, θ from the $4N$ equilibrium conditions

  Damp each target toward its prediction and record all gaps

end while

Compute U and annual land rent using (13)

Ensure: Equilibrium with fixed H and endogenous U

Algorithm 3 Open-city equilibrium

Require: $a, b, \varphi, K, \tau, \bar{U}$ and parameters

Initialize H, w, q, Q, θ

while a local target fails or $U \neq \bar{U}$ **do**

  Execute the closed-city equilibrium mapping for current H

  Compute U and predict population using (34)

  Apply a damped population update

end while

Compute annual land rent using (13)

Ensure: Equilibrium with fixed $\bar{U}$ and endogenous H

8.5 Fixed-distribution accounting benchmark

The toolkit also reports a deliberately constrained benchmark in which the baseline residence–workplace assignment is held fixed. This exercise is designed to isolate gains available without spatial reallocation. It is *not* a third market-clearing equilibrium closure.

Let a superscript F denote the fixed-distribution counterfactual. The benchmark holds fixed

$$H^F = H^0, \qquad H^F_{Mi} = H^0_{Mi}, \qquad H^F_{Ri} = H^0_{Ri}, \qquad \pi^F_{ij} = \pi^0_{ij}. \tag{35}$$

Thus neither the number of modeled workers, their workplace and residence distributions, nor their bilateral origin–destination assignments change. Residential and commercial floor-space rents and the commercial land-use share are also held at baseline values:

$$Q^F_i = Q^0_i, \qquad q^F_i = q^0_i, \qquad \theta^F_i = \theta^0_i. \tag{36}$$

For a pure transport experiment, φ and K are unchanged, so total, commercial, and residential floor space are fixed as well.

Fixed local quantities do *not* fix endogenous spillovers. Both effective density kernels depend on bilateral travel times. The benchmark therefore recomputes

$$\Upsilon^F_i = \sum_s e^{-\delta \tau^1_{is}} \frac{H^0_{Ms}}{K_s}, \qquad A^F_i = a^1_i (\Upsilon^F_i)^\lambda, \tag{37}$$

$$\Omega^F_i = \sum_r e^{-\rho \tau^1_{ir}} \frac{H^0_{Rr}}{K_r}, \qquad B^F_i = b^1_i (\Omega^F_i)^\eta. \tag{38}$$

Shorter travel times can consequently raise effective workplace and residential density even though the underlying employment and population vectors do not move.

With workplace labour and commercial floor-space inputs fixed, counterfactual output and the competitive wage are

$$Y^F_i = A^F_i (H^0_{Mi})^\alpha (\theta^0_i L^0_i)^{1-\alpha}, \qquad w^F_i = \frac{\alpha Y^F_i}{H^0_{Mi}}. \tag{39}$$

Wages are therefore *not* fixed: productivity-spillover gains pass into output and wages. The benchmark recomputes resident labour income using the fixed OD probabilities and w^F, but it does not use that income to clear the residential floor-space market because $Q^F = Q^0$ is maintained.

For a worker who remains on baseline route (i, j), baseline floor-space rents cancel from the proportional indirect-utility change. The log change is

$$\Delta \log u_{ij}^F = \log \frac{B_i^F}{B_i^0} + \log \frac{w_j^F}{w_j^0} - \kappa(\tau_{ij}^1 - \tau_{ij}^0). \tag{40}$$

The aggregate fixed-distribution log-welfare change is the baseline-flow weighted average

$$g^F = \sum_i \sum_j \pi_{ij}^0 \Delta \log u_{ij}^F = g_B^F + g_w^F + g_\tau^F, \tag{41}$$

where

$$g_B^F = \sum_{ij} \pi_{ij}^0 \log(B_i^F / B_i^0), \tag{42}$$

$$g_w^F = \sum_{ij} \pi_{ij}^0 \log(w_j^F / w_j^0), \tag{43}$$

$$g_\tau^F = -\kappa \sum_{ij} \pi_{ij}^0 (\tau_{ij}^1 - \tau_{ij}^0). \tag{44}$$

The table reports total worker welfare as $100[\exp(g^F) - 1]$ and each component as $100[\exp(g_k^F) - 1]$. Because the components combine additively in logs but multiplicatively in percentage levels, the displayed component percentages need not sum exactly to the displayed total. Fixed-distribution GDP is

$$100 \left(\frac{\sum_i Y_i^F}{\sum_i Y_i^0} - 1 \right). \tag{45}$$

For a pure transport experiment, annual land rent is unchanged by construction: q, Q, θ, and L are all fixed in (13). Productivity gains accrue to wages in (39); amenity and commuting gains accrue to workers in (40). The zero land-rent change is therefore an assumption of this partial-equilibrium benchmark, not a prediction that landowners cannot benefit in the closed- or open-city equilibria. Allowing rents to adjust while imposing (35) would generally violate one or more housing-market, firm, and mixed-use land-arbitrage conditions unless additional location-specific wedges were introduced.

8.6 Zero-spillover versions of all three scenarios

When `project.run_no_spillover_comparison` is enabled, the toolkit also runs closed-city, open-city, and fixed-distribution versions with

$$\lambda = 0, \qquad \eta = 0. \tag{46}$$

This switch removes the endogenous productivity response to employment effective density and the endogenous amenity response to residential effective density. The decay parameters δ and ρ remain in the parameter structure but are economically inactive because the corresponding elasticities are zero. Fundamentals, transport times, preferences, production, floor-space supply, commuting choice, and the applicable closure condition remain in the model.

The zero-spillover run is not obtained by merely setting two parameters to zero after loading the main inversion. It re-inverts the observed baseline under (46) and then solves the baseline and counterfactual under that same specification. This makes each reported percentage change an internally consistent comparison and ensures that differences between the two specifications reflect spillovers rather than mismatched baseline primitives. The main results are unchanged and remain in `outputs/simulation`. The sensitivity outputs are stored separately in `outputs/no_spillovers/simulation`; the combined comparison is written to `outputs/simulation/aggregate_changes_spillover_comparison.csv`.

Interpretation differs by scenario:

- In the **closed city without spillovers**, population remains fixed, but workers may relocate, wages, rents, and land use may adjust, and utility changes. The difference from the main closed-city result measures the contribution of endogenous productivity and amenity feedbacks, including their interaction with relocation.

- In the **open city without spillovers**, utility remains fixed and population adjusts. Migration, commuting resorting, wages, rents, land use, GDP, and land rent remain endogenous. The difference from the main open-city result shows how spillovers affect the city's population and capitalization response.

- In the **fixed distribution without spillovers**, OD assignments, population, employment, rents, and land use remain fixed. With no direct productivity or amenity shock, productivity, amenities, wages, GDP, and land rent therefore remain unchanged. Worker welfare changes only through the commuting-cost term g_τ^F. This is the cleanest model-based measure of the benefit of shorter travel times for baseline commuters.

The difference between the with- and without-spillover results is a sensitivity comparison, not in general an additive "spillover component." In the two equilibrium closures, spillovers change sorting, prices, land use, and (in the open city) population, so interactions are part of the difference. Only the fixed-distribution decomposition in (41) cleanly separates commuting, productivity/wage, and amenity terms in log welfare.

9 Parameter choices

Table 3: Current demonstration defaults in `project_config.yaml`

Parameter	Configuration key	Default	Interpretation
α	`production_share_labor`	0.80	Labour share in production
β	`expenditure_share_consumption`	0.75	Numeraire expenditure share
ϵ	`commuting_elasticity`	6.6941	Frechet shape/commuting elasticity
κ	`commuting_time_coefficient`	0.014744	Cost per travel-time unit
λ	`productivity_spillover`	0.0710	Productivity feedback
δ	`productivity_spatial_decay`	0.3617	Productivity decay per time unit
η	`amenity_spillover`	0.1553	Amenity feedback
ρ	`amenity_spatial_decay`	0.7595	Amenity decay per time unit
μ	`construction_capital_share`	0.75	Non-land development share
$1-\mu$	`construction_land_share`	0.25	Land share

These defaults reproduce the ARSW-style demonstration parameterization; they are not universal constants. For a new city, the user should document the source and empirical setting for every parameter. In particular:

- κ, δ, and ρ must be compatible with the unit of the travel-time matrices; changing minutes to hours without rescaling them changes the model.

- ϵ governs both sorting responses and the open-city population update, so welfare and migration results can be sensitive to it.

- λ and η govern amplification through endogenous productivity and amenity. Zero is the transparent no-spillover sensitivity case.

- α, β, and $1-\mu$ determine the incidence of revenue across labour, floor-space uses, and land. In particular, annual land rent in (13) uses $1-\mu = 0.25$.
- Numerical tolerances and damping weights are algorithmic controls, not economic parameters; changing them should affect convergence, not the converged equilibrium.

10 Counterfactual outputs and validation

For each requested closure, the toolkit solves both τ^0 and τ^1, computes block-level percentage changes, and reports aggregate utility, population, GDP, and annual land rent. The main aggregate table uses

$$\%\Delta GDP = 100\left(\frac{\sum_i Y_i^1}{\sum_i Y_i^0} - 1\right), \tag{47}$$

$$\%\Delta\mathcal{R} = 100\left(\frac{\sum_i (1-\mu)[q_i^1\theta_i^1 L_i + Q_i^1(1-\theta_i^1)L_i]}{\sum_i (1-\mu)[q_i^0\theta_i^0 L_i + Q_i^0(1-\theta_i^0)L_i]} - 1\right). \tag{48}$$

For a pure transport counterfactual φ and K are fixed, hence L is fixed, but q, Q, and θ need not be. Consequently, percentage changes in annual land rent are not generally equal to changes in either floor-space rent.

10.1 Three travel-time measures

Let π^0 and π^1 denote baseline and counterfactual unconditional OD probabilities, τ^0 and τ^1 the corresponding travel-time matrices, and H^0, H^1 modeled commuter populations. Define

$$\bar{T}_{00} = \sum_{ij} \pi_{ij}^0 \tau_{ij}^0, \qquad \bar{T}_{01} = \sum_{ij} \pi_{ij}^0 \tau_{ij}^1, \qquad \bar{T}_{11} = \sum_{ij} \pi_{ij}^1 \tau_{ij}^1. \tag{49}$$

The aggregate output table distinguishes:

Mean commute time, baseline flows `MeanCommuteTimePct_BaselineFlows` is $100(\bar{T}_{01}/\bar{T}_{00} - 1)$. It changes the network while retaining the baseline OD weights. It is the immediate or before-relocation network effect for the original commuting pattern.

Mean commute time, counterfactual flows `MeanCommuteTimePct_CounterfactualFlows` is $100(\bar{T}_{11}/\bar{T}_{00} - 1)$. It uses post-adjustment OD weights and hence includes the effect of changed residence–workplace patterns on the average commute.

Aggregate commuter-minutes `AggregateCommuterMinutesPct` is $100[H^1\bar{T}_{11}/(H^0\bar{T}_{00}) - 1]$. It measures the total one-way minutes across all modeled commuters. It equals the counterfactual-flow mean measure when population is fixed, but differs in the open city because H^1 changes.

All three measures use one-way commuting time. They are physical time statistics, not monetized benefits. The fixed-distribution benchmark has $\pi^1 = \pi^0$ and $H^1 = H^0$, so all three percentage changes coincide.

Before interpretation, users should verify:

1. location IDs and matrix ordering are identical;
2. employment and population aggregate to the same number;
3. inversion reproduces both observed employment margins;
4. all fixed-point gaps satisfy the configured tolerance;
5. baseline equilibrium reproduction is satisfactory;
6. land-rent output is constructed from (13), not from an input field labelled land value.

11 Code correspondence

File	Economic role
`invert_baseline.m`	Commented baseline-inversion entry script.
`run_counterfactual.m`	Solves the requested equilibrium closures and the fixed-distribution benchmark.
`run_no_spillovers.m`	Re-inverts and reruns all three scenarios with $\lambda = \eta = 0$, saving results separately.
`qt_invert_baseline.m`	Passes the single observed rent into both use-specific inversion conditions.
`cmodexog.m`	Inverts A, B, w and commuting probabilities jointly.
`cdensityE.m`	Inverts $L_M, L_R, L, \theta, \varphi$.
`cprod.m`, `cres.m`	Recover a and b from total productivity and amenity.
`ubar.m`	Computes baseline expected utility $\bar{U}$.
`smodendog.m`	Solves the closed-city $4N$-equation target system.
`ussmodendog.m`	Adds population and utility clearing for the open city.
`qt_solve_fixed_distribution.m`	Evaluates fixed OD assignments with transport-mediated spillovers and decomposes worker welfare.
`qt_write_closure_results.m`	Reports local outcomes, annual land rent, and the three travel-time measures.
`qt_display_results.m`	Displays compact MATLAB headings while preserving descriptive CSV variable names.

12 Interpretation and limitations

Results are comparative statics conditional on the model, parameters, baseline data, and travel times. Wages are always recovered inside the model inversion; they are never generated from an auxiliary employment elasticity. Synthetic rents are useful for testing the pipeline but not for policy valuation. Observed rents require compatible units, concepts, and dates. Annual land rent is a model-implied flow; converting it to a land asset value requires an explicit discounting and expectations model. Open-city utility is fixed by construction, so population rather than utility is its aggregate adjustment margin.

Reference

Ahlfeldt, G. M., Redding, S. J., Sturm, D. M., and Wolf, N. (2015). "The Economics of Density: Evidence from the Berlin Wall." *Econometrica*, 83(6), 2127–2189. doi:10.3982/ECTA10876.